

Internship Project Report | Alfido Tech

1. Project Objective

The objective of this project is to implement core data structures from scratch using C++. The project focuses on understanding insertion, deletion, and searching operations in data structures.

2. Technologies & Modules Used

Language: C++

Development Environment: Visual Studio Code (VS Code)

Data Structures Implemented:

- *Linked List*
- *Stack*

3. Project Description

This project demonstrates the implementation of Linked List and Stack using C++.

The following operations were implemented:

Linked List

- *Insert Node*
- *Delete Node*
- *Search Node*

```
main.cpp
1 #include <iostream>
2 using namespace std;
3
4 struct Node{
5     int data;
6     Node* next;
7 };
8
9 Node* head = NULL;
10
11 void insert(int x){
12     Node* n = new Node();
13     n->data = x;
14     n->next = head;
15     head = n;
16 }
17
18 void search(int x){
19     Node* temp = head;
20     while(temp){
21         if(temp->data == x){
22             cout << "\nElement Found";
23             return;
24         }
25         temp = temp->next;
26     }
27
28     cout << "\nElement Not Found";
29 }
30
31 void deleteNode(int x){
32     Node *temp = head, *prev = NULL;
33     while(temp && temp->data != x){
34         prev = temp;
35         temp = temp->next;
36     }
37
38     if(temp == NULL) return;
39
40     if(prev == NULL)
41         head = temp->next;
42     else
43         prev->next = temp->next;
44
45     delete temp;
46 }
47
48 void display(){
49     Node* temp = head;
50     while(temp){
51         cout << temp->data << " ";
52         temp = temp->next;
53     }
54 }
```

```
main.cpp
47
48 void display(){
49     Node* temp = head;
50     while(temp){
51         cout << temp->data << " ";
52         temp = temp->next;
53     }
54 }
55
56 int main(){
57     insert(10);
58     insert(20);
59     insert(30);
60
61     cout << "Linked List: ";
62     display();
63
64     search(20);
65
66     deleteNode(20);
67
68     cout << "\nAfter Delete: ";
69     display();
70
71     return 0;
72 }
```

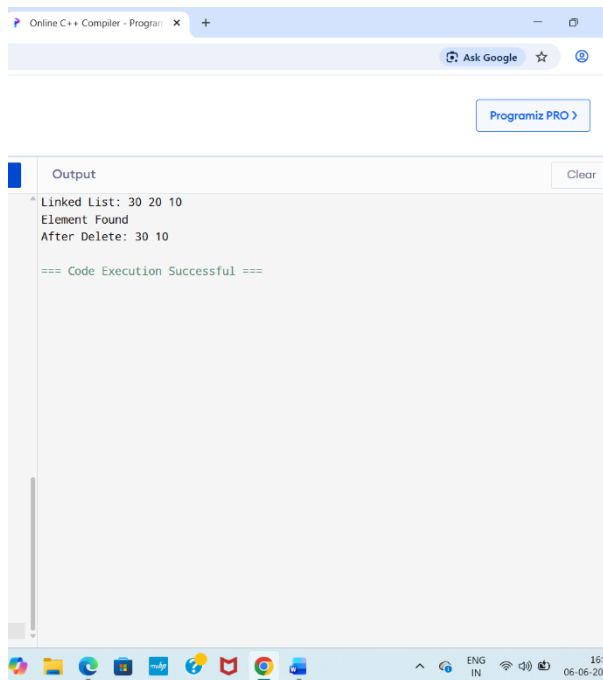
Complexity Analysis

Linked List

- *Insert* : $O(1)$
- *Delete* : $O(n)$
- *Search* : $O(n)$

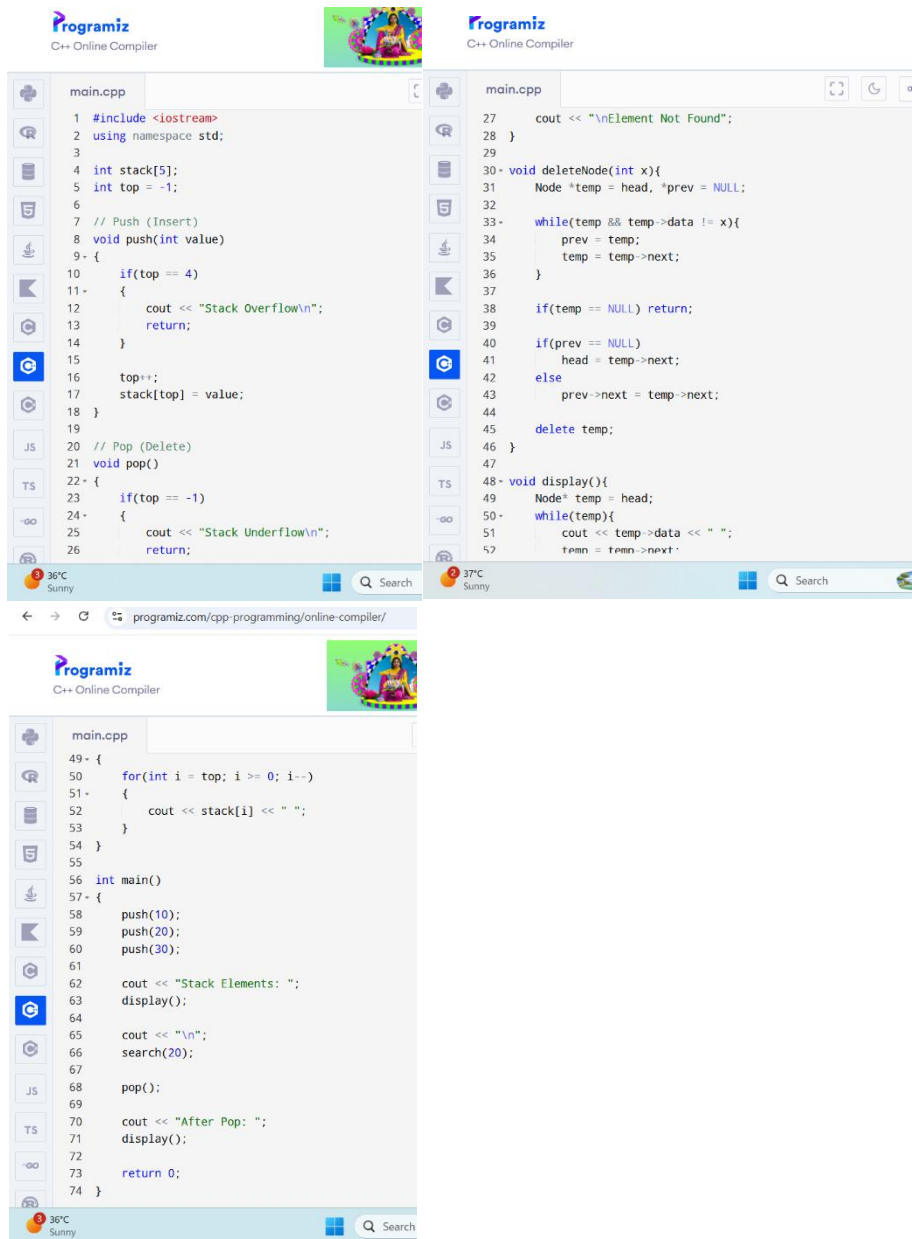
Linked List Output

Linked List: 30 20 10
Element Found
After Delete: 30 10



Stack

- *Push Element*
- *Pop Element*
- *Search Element*



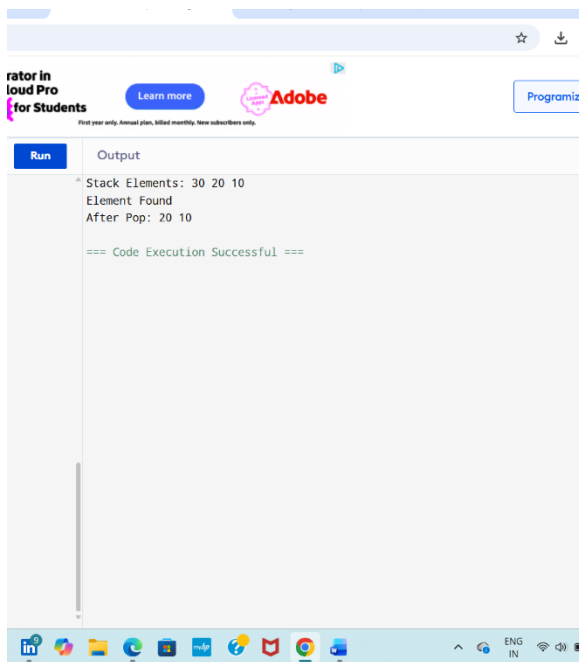
Complexity Analysis

Stack

- *Push* : $O(1)$
- *Pop* : $O(1)$
- *Search* : $O(n)$

Stack Output

Stack: 30 20 10
Element Found
After Pop: 20 10



4. Key Learning Outcomes

- *Learned implementation of Linked List and Stack.*
- *Improved understanding of Data Structures.*
- *Enhanced problem-solving and debugging skills.*
- *Gained practical experience in C++ programming.*
- *Understood complexity analysis of operations.*

5. Conclusion

This project provided practical experience in implementing fundamental data structures using C++. It strengthened my understanding of Linked Lists, Stacks, and their operations while improving my programming and analytical skills.